

The left edge of Rule 30 without Rule 30: the stripe recursion, Wolfram’s number 2,107,985,255, and the forks beyond it

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Abstract

The left edge of the Rule 30 pattern grown from one black cell is made of diagonals that become periodic, with periods that are powers of two. Wolfram tabulated the depths at which the period first reaches 2, 4, 8, 16, 32 and gave the sixth, where it reaches 64, as “2,107,985,255 or more”. The eventual patterns of consecutive diagonals obey a two-term recursion with a closed form, which we run without simulating the rule; it is deterministic except at a diagonal that ends all white, where it forks. The eighth such diagonal of the seed is $d = 1,420,878,968$, a genuine fork: on one branch the period reaches 64 at exactly $d = 2,107,985,255$, on the other not below $3.34 \cdot 10^9$, and beyond the doubling the first branch has no further fork up to 10^{12} . Over more than 10^{10} pairs, consecutive patterns differ in a binomially distributed number of cells; if this holds down to distance 0, forks come about once per 2^P diagonals at period P and the next doubling lies near 10^{19} – 10^{20} , beyond any computation. We prove that at a fork the order–chaos front advances exactly when the frozen diagonal is black on the row after the previous front, and that, given the front’s path over the three preceding diagonals, five cells of the last three patterns force the seed onto the branch of Wolfram’s number. That path is the one input not proved; it is the same in all eight coin-driven models of the front whose path we recorded across the fork, and all twenty-eight models we ran take that branch. The centre column the stripes alone would produce passes every fair-coin test we applied to 10^9 bits but one, and the real centre column is that sequence XOR a transient bit that behaves, in every test we ran, as an independent coin.

1 Introduction

Rule 30 started from a single black cell produces a pattern whose right side looks random and whose left side is striped: each diagonal parallel to the left edge becomes periodic after a transient, with a period that is a power of two [2, 3, 4]. The period of the stripes grows. Wolfram [5, p. 871] lists the depths at which the maximal period first reaches 2, 4, 8, 16, 32 as 3, 8, 29, 400, 87,867 and the sixth, period 64, as “2,107,985,255 or more”. Gravner and Griffeath [2, §8] describe the eventual patterns as the labels of a tree and record that “the first genuine branching occurs for period 16 at generation 53,207”.

This note does four things.

*Computational and editorial assistance from Claude (Anthropic). Every numbered result carries a status label, defined in Section 1; no claim is made beyond its label.

1. It writes the recursion between consecutive eventual patterns in closed form (Lemma 3.1) and runs it forward in C at about $3 \cdot 10^7$ diagonals a second on one core, with no simulation of the rule. The recursion forks only at a diagonal that ends all white.
2. It computes the eighth such diagonal of the seed, $d^* = 1,420,878,968$, a genuine fork, and shows that one branch doubles the period at Wolfram's number to the digit while the other does not double below $3.34 \cdot 10^9$ (Theorem 4.1).
3. It gives the measured reason why a seventh doubling is out of reach: consecutive patterns differ binomially, so forks come about once per 2^P diagonals at period P , which matches the spacing of the doublings (Law 5.1).
4. It shows what decides which branch the picture takes. At a diagonal that ends all white the front between the striped and the chaotic region advances exactly when that diagonal is black on the row after the previous front (Theorem 6.2), and in the cases met at the seed's forks this fixes the branch. For the seed at d^* a five-line derivation gives the branch of Wolfram's number from five pattern cells and the front's path modulo 32 over the three preceding diagonals (Theorem 6.5).

The one input not proved is that path, which every coin-driven model of the front that we recorded shares; we isolate it as the first open problem. The recursion itself is the forward form of the label tree of Gravner and Griffeath [2], and its closed form is the reset formula of [1], the closed form of any eventually periodic diagonal, applied to the eventual patterns; the parity rule at a frozen diagonal is Rowland's [4], and that the two branches at a doubling are translates is in [2, §8]. What is new here is the use of the recursion as a fast forward computation, and what it shows about the forks. The note is self-contained: the facts about the front that its proofs use are proved in Section 2. Our longer manuscript [1] is cited only for context. Results carry a status: PROVED; EXACT for a finite computation reproducible from the accompanying code; MEASURED for a statistical statement; KNOWN for the literature; OPEN.

2 Setting

Rule 30 is $u_x^{t+1} = u_{x-1}^t \oplus (u_x^t \vee u_{x+1}^t)$ on $x \in \mathbb{Z}$, from $u_x^0 = [x = 0]$. The *diagonal frame* is $D_d(t) = u_{d-t}^t$ for $d, t \geq 0$: diagonal $d = 0$ is the left edge $x = -t$, which is always black. In this frame the rule reads

$$D_{d+1}(t+1) = D_{d-1}(t) \oplus (D_d(t) \vee D_{d+1}(t)) \quad (d, t \geq 0, D_{-1} \equiv 0). \quad (1)$$

Every D_d is eventually periodic with minimal period p_d a power of two [3, 2, 4]. Let $r_d: \mathbb{N} \rightarrow \mathbb{F}_2$ be the periodic sequence with $D_d(t) = r_d(t)$ for all large t (the *label* of diagonal d), τ_d the least τ with $D_d(t) = r_d(t)$ for all $t \geq \tau$, and $\Delta_d = D_d \oplus r_d$ the *defect*, which has finite support ending at row $\tau_d - 1$ (empty if $\tau_d = 0$). The *front* is $M_d = \max(\tau_{d-1}, \tau_d) - 1$, the last row on which diagonal $d - 1$ or d is still transient. We say row t lies *beyond the front* of diagonal d if $t \geq M_d + 1$; there both $D_{d-1}(t) = r_{d-1}(t)$ and $D_d(t) = r_d(t)$. Diagonal d is *frozen* if $r_d \equiv 0$, that is, if it ends all white. The front *moves* (or advances) at d if $M_d > M_{d-1}$, and *stalls* at d otherwise. We call the sequence of labels the *wall*, and a sequence of labels generated by the recursion of Section 3, with a choice made at each fork, a *lineage*. Figure 1 shows the picture.

The diagonal $d = 0$ is black on every row: $u_{-t}^t = u_{-t-1}^{t-1} \oplus (u_{-t}^{t-1} \vee u_{-t+1}^{t-1}) = 0 \oplus (0 \vee 1) = 1$ by induction, so $r_0 \equiv 1$ and $\tau_0 = 0$. The rest of what the proofs below use about labels and fronts is the following.

Lemma 2.1 (labels and the front; PROVED). (a) Background equation. $r_{d+1}(t+1) = r_{d-1}(t) \oplus (r_d(t) \vee r_{d+1}(t))$ for all $d \geq 0$ and all t , with $r_{-1} \equiv 0$ and the labels extended periodically to all integers.

(b) No two consecutive labels are both zero.

(c) Monotonicity. $M_d \leq M_{d+1}$.

(d) Landing rule. Either $M_{d+1} = M_d$, or diagonal d is not frozen and $M_{d+1} = n := \min\{t \geq M_d + 1 : r_d(t) = 1\}$, the first black cell of r_d beyond the front; in the second case the defect of diagonal $d+1$ is 1 on the rows $M_d + 1, \dots, n$ and ends there, $\tau_{d+1} - 1 = M_{d+1} = n$.

(e) If diagonal d is frozen, $M_{d+1} = M_d$.

Proof. (a) For t beyond $\tau_{d-1}, \tau_d, \tau_{d+1}$ the three diagonals equal their labels and (1) reads as stated; both sides are periodic in t , so the identity holds for all t . (b) Suppose $r_{d-1} = r_d \equiv 0$ with $d \geq 1$. Then (a) with $d-1$ in place of d gives $0 = r_d(t+1) = r_{d-2}(t)$, so $r_{d-2} \equiv 0$ too, and descending, $r_0 \equiv 0$, which is false. (This is the argument of [2, Proposition 3.3].) (c), (d), (e) Subtracting (a) from (1) gives, for every t ,

$$\Delta_{d+1}(t+1) = \Delta_{d-1}(t) \oplus (D_d(t) \vee D_{d+1}(t)) \oplus (r_d(t) \vee r_{d+1}(t)).$$

If $t \geq \tau_d$ and $t \geq \tau_{d+1}$ the last two terms cancel, so $\Delta_{d-1}(t) = \Delta_{d+1}(t+1) = 0$ for $t \geq \max(\tau_d, \tau_{d+1})$; hence $\tau_{d-1} \leq \max(\tau_d, \tau_{d+1})$, which is (c). For $t \geq M_d + 1$ the first term vanishes and $D_d = r_d$, so, using $(x \vee a) \oplus (x \vee b) = \neg x \wedge (a \oplus b)$,

$$\Delta_{d+1}(t+1) = \neg r_d(t) \wedge \Delta_{d+1}(t) \quad (t \geq M_d + 1).$$

If $r_d \equiv 0$ the defect Δ_{d+1} is constant from row $M_d + 1$ on, hence 0 there, so $\tau_{d+1} \leq M_d + 1$ and, with (c), $M_{d+1} = M_d$: this is (e). Otherwise Δ_{d+1} keeps its value $\Delta_{d+1}(M_d + 1)$ on the rows $M_d + 1, \dots, n$, where r_d is white, and is 0 from row $n + 1$ on. If that value is 0 then $\tau_{d+1} \leq M_d + 1$ and $M_{d+1} = M_d$ by (c). If it is 1 then $\tau_{d+1} = n + 1$, so $M_{d+1} = \max(\tau_d, n + 1) - 1 = n$, since $n \geq M_d + 1 \geq \tau_d$. This is (d). $\square$

In every case examined, the labels of a left half of Rule 30, driven by its boundary column from a finite initial state, are the seed's up to a translation of rows: this is KNOWN for every finite seed up to diagonal 53,207 [2, §8], and EXACT for seven combinations of boundary column, some random, and initial state up to diagonal 12,000 [1]. We use this only as a measured hypothesis, in Proposition 6.3.

3 The stripe recursion

Lemma 3.1 (PROVED). Let $(a, b) = (r_{d-1}, r_d)$ and $c = r_{d+1}$. Then $c(t+1) = a(t) \oplus b(t) \oplus (\neg b(t) \wedge c(t))$ for all t .

(a) If $b \not\equiv 0$, extend a, b, c periodically to all integers and let $u^* < t$ be the last row before t with $b(u^*) = 1$. Then $c(t) = a(u^*) \oplus 1 \oplus \bigoplus_{u^* < s < t} a(s)$. So c is the unique periodic solution, and its period divides $\text{lcm}(p_{d-1}, p_d)$.

(b) If $b \equiv 0$ (diagonal d is frozen), $c(t) = c(0) \oplus \bigoplus_{0 \leq s < t} a(s)$ for $t \geq 0$. Both solutions are periodic, and they are complements. Their minimal period is p_{d-1} if a has even weight on a period and $2p_{d-1}$ if odd.

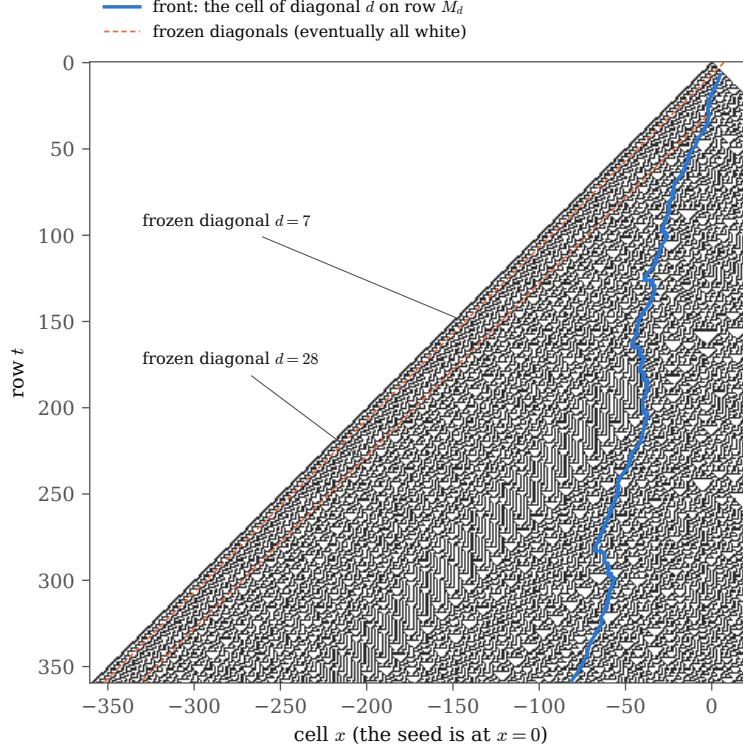


Figure 1: The left edge of Rule 30 from one cell, rows 0–359. Diagonals run from upper right to lower left. Left of the blue front each diagonal is periodic (the stripes); right of it the pattern is still transient. The frozen diagonals $d = 7$ and $d = 28$ are all white from rows 0 and 31 on (dashed); $d = 2$, white from row 2 on, runs along the edge and is not marked. Here the front leans left by about 0.2 cells per row; over the first 800,000 diagonals its average lean is 0.24.

Proof. The recursion is Lemma 2.1(a), since $b \vee c = b \oplus (\neg b \wedge c)$. Where $b(s) = 1$ the recursion gives $c(s+1) = a(s) \oplus 1$ whatever $c(s)$ was; where $b(s) = 0$ it gives $c(s+1) = a(s) \oplus c(s)$. Iterating from u^* gives (a). Every solution agrees with (a) after the first black cell of b , so the periodic one is unique and inherits every common period of a and b . In (b), $c(t+p) = c(t) \oplus \bigoplus_{s=t}^{t+p-1} a(s)$ for $p = p_{d-1}$, which is $c(t)$ at even weight and $\neg c(t)$ at odd weight. Since $a(t) = c(t) \oplus c(t+1)$, every period of c is a period of a , so the minimal period of c is p at even weight and $2p$ at odd weight. $\square$

Lemma 3.1 is that reset formula applied to the labels, and part (b) is Rowland's parity law [4]. The recursion also runs backwards: $a(t) = c(t+1) \oplus b(t) \oplus (\neg b(t) \wedge c(t))$ recovers r_{d-1} from (r_d, r_{d+1}) , so each pair of consecutive labels has exactly one predecessor pair, and this *backward map* commutes with translation of rows. Read backwards, the wall is the tree of Gravner and Griffeath [2].

Corollary 3.2 (PROVED). (i) Diagonal $d \geq 2$ is frozen if and only if $r_{d-2} = r_{d-1}$: forks are twins. (ii) The running maximum of p_d grows only across a frozen diagonal, from $d-1$ to $d+1$, and then by a factor of two. (iii) At a frozen diagonal whose predecessor label has odd weight, the two candidates are translates of each other by p_{d-1} rows. At each frozen diagonal of Table 1 where it has even weight, no translation maps one candidate to the other, so the branches are different walls.

Proof. (i) Put $c \equiv 0$ in the recursion: $a = b$. Conversely if $a = b \neq 0$ then (a) gives $c(t) = a(u^*) \oplus 1 \oplus 0 = 0$; and $a = b \equiv 0$ cannot occur by Lemma 2.1(b). (ii) By Lemma 3.1(a), $p_{d+1} \leq \max(p_{d-1}, p_d)$

frozen diagonal d	period of r_{d-1}	weight	type	new period, first at
2	1	1	doubling	2 at $d = 3$
7	2	1	doubling	4 at $d = 8$
28	4	3	doubling	8 at $d = 29$
399	8	3	doubling	16 at $d = 400$
53,207	16	6	genuine fork	
58,286	16	10	genuine fork	
87,866	16	5	doubling	32 at $d = 87,867$
1,420,878,968	32	20	genuine fork	
2,107,985,254 ^a	32	17	doubling	64 at $d = 2,107,985,255$
3,340,408,059 ^b	32	16	genuine fork	

Table 1: Frozen diagonals of the seed’s lineage (EXACT), with the period and weight of r_{d-1} . Odd weight gives a doubling, even weight a genuine fork. The branches at $d^* = 1,420,878,968$ are 93425cac and 6cbda353; ^aon the branch 6cbda353, ^bon the branch 93425cac.

unless d is frozen; if it is, $p_{d+1} \leq 2p_{d-1}$ by (b), and $2p_{d-1}$ exceeds the running maximum only if p_{d-1} equals it. (iii) At odd weight $c(t + p_{d-1}) = \neg c(t)$, and $\neg c$ is the other candidate. At the even-weight frozen diagonals of Table 1 the candidate weights are 4/12 at 53,207, 6/10 at 58,286 and 14/18 at d^* , so they are not translates; at 3,340,408,059 both weigh 16, and a direct check of the 32 rotations finds none that maps one to the other. No later pair of consecutive labels on one branch is a translate of the pair of the same index on the other either: the backward map, which recovers (r_{j-1}, r_j) from (r_j, r_{j+1}) , commutes with translation, so such a pair would make the candidates translates. $\square$

The first statement of (iii) is in [2, §8]. Every frozen diagonal is a fork of the recursion; we call it a *doubling* when r_{d-1} has odd weight, since the period then doubles and the two branches are one wall up to translation, and a *genuine fork* when the weight is even.

Validation (EXACT). Started from (r_{19}, r_{20}) and given the seed’s branch at each fork, the recursion reproduces the labels of a direct simulation at all 799,980 diagonals up to 800,000. Implementations with 64- and 128-bit pattern words, run from $d = 90,000$, stop at the same frozen diagonal d^* with the same labels and the same count of diagonals of each period.

4 The eighth frozen diagonal and Wolfram’s number

Theorem 4.1 (EXACT). *The frozen diagonals of the seed’s lineage up to the fork, and on each branch after it up to the next frozen diagonal, are those of Table 1. The eighth, $d^* = 1,420,878,968$, is a genuine fork: r_{d^*-1} has period 32 and weight 20, and the candidates for r_{d^*+1} are 93425cac and 6cbda353 (bit t of the 32-bit word is the cell on row t modulo 32). On the branch 6cbda353 the next frozen diagonal is 2,107,985,254, of odd weight 17, and the period first reaches 64 at $d = 2,107,985,255$. On the branch 93425cac the next frozen diagonal is 3,340,408,059, of even weight 16, and the period stays at most 32 up to it.*

Remark 4.2. Wolfram’s “2,107,985,255 or more” is consistent with a computation of this kind that met the fork at d^* and reported the earlier of its two outcomes as a lower bound. We do not claim to know how the number was obtained.

Theorem 4.3 (EXACT). *Beyond 2,107,985,255 the lineage of the branch 6cbda353 has no frozen diagonal up to $d = 10^{12}$, so its period-64 regime lasts at least until $d = 10^{12}$. The computation*

beyond $1.508 \cdot 10^{11}$ ran on a second machine, from a checkpoint of the first, in 8,492 steps of 10^8 diagonals. Thirteen of them, at six places up to $6.503 \cdot 10^{11}$, were recomputed on the first machine by two separately written programs, with 64- and 128-bit pattern words, and agree to the last hex digit.

The statement does not depend on the branch taken at 2,107,985,254. That frozen diagonal has odd weight, so its two candidates are translates (Corollary 3.2(iii)). By the uniqueness in Lemma 3.1(a) the two lineages after it are then translates of each other, and being frozen is invariant under translation. Inside the regime a label occasionally has minimal period 32, as Lemma 3.1(a) allows: 188 of the $8.492 \cdot 10^{11}$ labels computed on the second machine, against 197.7 ± 14.1 expected for uniformly random 64-bit patterns.

5 The fork law

Law 5.1 (MEASURED). *In a regime of period P , the number of rows in a period where consecutive labels r_d and r_{d+1} differ is distributed as $\text{Binomial}(P, \frac{1}{2})$.*

Figure 2 shows the evidence. Over the $1.42 \cdot 10^9$ consecutive pairs of the seed's period-32 regime from $d = 90,000$ to the fork, the counts fit the binomial with $\chi^2 = 29.7$ on the 31 classes with expectation at least 5 (distances 1–31); at distances 0 and 32, where 0.33 pairs are expected, 1 and 0 are seen. Over the first 10^{10} pairs of the period-64 regime they fit it with $\chi^2 = 51.7$ on the 47 classes with expectation at least 5, and on both tails: at distance 8, 2.4 pairs are expected and 3 seen; at 56 and 57, 2.4 and 0.3 are expected and 1 and 1 seen.

By Corollary 3.2(i) a frozen diagonal is the distance-0 class, so if the law holds down to distance 0, frozen diagonals come at rate 2^{-P} ; the single pair at distance 0 against 0.33 expected at $P = 32$ is all the direct evidence at that distance. Heuristically, if about half of them have odd weight, as for independent uniform patterns, about half end the regime, and the expected length of a period- P regime is about 2^{P+1} diagonals: $1.3 \cdot 10^5$ at $P = 16$, $8.6 \cdot 10^9$ at $P = 32$, $3.7 \cdot 10^{19}$ at $P = 64$. The seed's period-16 regime is 87,467 long with three frozen diagonals; on the branch of Wolfram's number its period-32 regime is $2.1 \cdot 10^9$ long with two. Both are within Poisson scatter of these values. So a period of 128 is expected near 10^{19} – 10^{20} diagonals, and if the seed takes that branch (Theorem 6.5), 2,107,985,255 is the last doubling of its left edge within reach of computation.

The law is the wall's own pseudo-randomness statement: the orbit of the recursion on the 4^P label pairs looks, in the distances between consecutive labels, like a sequence of independent uniform patterns. What is proved is only its link to the forks, Corollary 3.2(i). We know no route to a proof of the law itself, which would need a mixing property of the recursion's orbit.

6 Which branch the picture takes

Theorem 6.1 (branch rule; PROVED). *Let d be frozen with $\tau_d \geq 1$, so $D_d(t) = 0$ for $t \geq \tau_d$ and $D_d(\tau_d - 1) = 1$. Then $D_{d+1}(\tau_d) = D_{d-1}(\tau_d - 1) \oplus 1$ and $D_{d+1}(t+1) = D_{d-1}(t) \oplus D_{d+1}(t)$ for $t \geq \tau_d$. The label taken is the candidate with*

$$r_{d+1}(\tau_d - 1) = 1 \oplus \bigoplus_{t \geq \tau_d - 1} \Delta_{d-1}(t).$$

The branch is fixed by the cells of diagonal $d - 1$ from row $\tau_d - 1$ on.

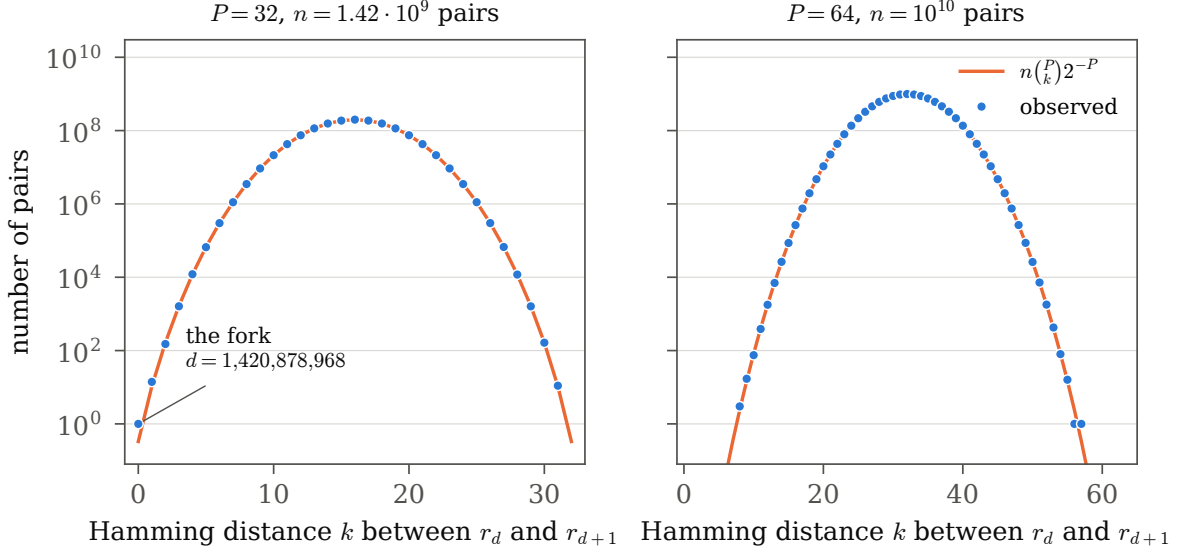


Figure 2: The fork law. Number of consecutive label pairs at Hamming distance k (dots) against $n \binom{P}{k} 2^{-P}$ (line). Left: the period-32 regime of the seed from $d = 90,000$ to the fork; the single pair at $k = 0$ is the fork d^* . Right: the first 10^{10} pairs of the period-64 regime.

Proof. Apply (1) at row $\tau_d - 1$, where $D_d = 1$ forces the OR to 1, and at rows $t \geq \tau_d$, where $D_d = 0$ leaves $D_{d+1}(t)$. Both candidates c satisfy $c(t+1) = r_{d-1}(t) \oplus c(t)$. Subtracting, $\Delta_{d+1}(t+1) = \Delta_{d-1}(t) \oplus \Delta_{d+1}(t)$ for $t \geq \tau_d$, so Δ_{d+1} is eventually the constant $\Delta_{d+1}(\tau_d) \oplus \bigoplus_{t \geq \tau_d} \Delta_{d-1}(t)$, which must vanish. Expanding $\Delta_{d+1}(\tau_d)$ with the first identity and the candidate recursion at row $\tau_d - 1$ gives the displayed condition. $\square$

Theorem 6.2 (advance criterion; PROVED). *Let $d \geq 2$ be frozen and $m = M_{d-1}$. From row $m+1$ on, D_d keeps its value until the first black cell of r_{d-1} at or after $m+1$, and is white after it. Hence the front advances at d if and only if $D_d(m+1) = 1$. It then lands on $\min\{t \geq m+1 : r_{d-1}(t) = 1\}$ and the branch taken has $r_{d+1}(M_d+1) = 0$. Otherwise $M_d = m$. If the front moved at $d-1$, the criterion reads: the front stalls at d if and only if $D_d(m) = 1$, and then the branch taken has $r_{d+1}(m) = 0$. In that case a black cell of r_{d-3} strictly between the last two landing rows M_{d-2} and m forces the stall.*

Proof. By Corollary 3.2(i), $r_{d-2} = r_{d-1} =: r$. For $t \geq m+1$ both $D_{d-2}(t)$ and $D_{d-1}(t)$ equal $r(t)$, since t lies beyond the front $M_{d-1} = m$. So (1) gives $D_d(t+1) = r(t) \oplus (r(t) \vee D_d(t))$, which is 0 where $r(t) = 1$ and $D_d(t)$ where $r(t) = 0$. This proves the first sentence. Since $r_d \equiv 0$, $\tau_d - 1$ is the last black row of D_d . If $D_d(m+1) = 1$ that row is the first black cell n of r at or after $m+1$, so $M_d = n$. At row n , $D_{d+1}(n+1) = D_{d-1}(n) \oplus (D_d(n) \vee D_{d+1}(n)) = 1 \oplus 1 = 0$, and row $n+1$ lies beyond the front of $d+1$, which is $M_{d+1} = M_d = n$. If $D_d(m+1) = 0$ then D_d is white from $m+1$ on, so $\tau_d \leq m+1$ and $M_d = m$. After a move at $d-1$, the landing row m is a black cell of $r_{d-2} = r$ and carries the defect of diagonal $d-1$ (Lemma 2.1(d)), so $D_{d-1}(m) = \neg r(m) = 0$ and $D_{d-2}(m) = r(m) = 1$. So $D_d(m+1) = 1 \oplus D_d(m)$. At a stall $D_d(m) = 1$ and D_d is white from $m+1$ on, so $\tau_d = m+1$, and Theorem 6.1 gives $r_{d+1}(m) = 1 \oplus \Delta_{d-1}(m) = 0$, since Δ_{d-1} is 1 at its landing row m and 0 beyond it. On the window $W = (M_{d-2}, m)$ the label r is white, because m is the first black cell after M_{d-2} . There $D_d(s+1) = D_{d-1}(s) \vee D_d(s)$, so D_d never decreases on W , while $D_{d-1}(s+1) = r_{d-3}(s) \oplus D_{d-1}(s)$. Counting back from $D_{d-1}(m) = 0$, the last black cell u of r_{d-3} in W gives $D_{d-1}(u) = 1$, hence $D_d(u+1) = 1$ and $D_d(m) = 1$. $\square$

Check (EXACT). At six of the seven frozen diagonals of the seed up to 87,866 the front either advances (2, 28 and 58,286) or stalls right after a move (399, 53,207 and 87,866). In all six the branch given by Theorem 6.2 is the seed's, on a direct simulation of the cells. The remaining one, $d = 7$, is white from row 0 on ($\tau_7 = 0$), and the front moves neither there nor at $d = 6$, so neither case applies.

Proposition 6.3 (phase congruence; PROVED with an EXACT step; its hypotheses MEASURED). *Call a left machine the cells $x \leq -1$ evolved by the rule from a finite initial state, with column 0 replaced by a given boundary sequence; it has its own diagonal frame, labels, defects and front, defined as above but with diagonals indexed from its own left edge (the leftmost black cell, which recedes one cell per row), and the seed's left half is the left machine driven by the seed's own centre column. Suppose a left machine has on a stretch of diagonals containing 20,000–87,866 or 90,000–800,000 the seed's labels translated by s_w rows and the seed's fronts translated by s_b rows, and that its front stays left of its boundary column there. Then $s_b \equiv s_w$ modulo the period of that regime, 16 or 32. If the stretch ends at a frozen diagonal where the seed's front advances, or stalls right after a move, the machine takes the seed's branch there translated by s_b rows; at even weight this is the translation by s_w .*

Proof. The landing rule holds for the machine. Its proof, Lemma 2.1(d), uses the background equation, the definitions, and (1) only on rows beyond the front. For the machine (1) holds at every cell left of its boundary column, which includes those rows by hypothesis, and the background equation holds because every diagonal is periodic far down, where it lies in the left half. Written in the seed's rows, the machine's landing rule says that the seed's landing rule holds shifted by $s = s_b - s_w$ at every diagonal where the front moves. EXACT: the shifted rule holds for $s \equiv 0$ at all 37,086 moves of the period-16 regime between diagonals 20,000 and 87,866, and at all 385,784 moves of the period-32 regime between diagonals 90,000 and 800,000, and fails for every other residue; the best other residue satisfies it at 26.1% of the moves in both regimes. The branch then follows from Theorem 6.2: the selected candidate is the one white on a row fixed relative to the front, $M_d + 1$ after an advance and m after a stall, which for the machine is that row translated by s_b . The machine's candidates are the seed's translated by s_w . At even weight they have the regime's period or a divisor of it, so the translations by s_w and s_b agree. At odd weight the two candidates are translates of each other by p_{d-1} , and the one white on the machine's row is the seed's choice translated by s_b . $\square$

Its hypotheses are the universality of the wall beyond period 8, KNOWN for every finite seed to diagonal 53,207 [2, §8] and EXACT for the seven driven machines of Section 2 to 12,000 [1], and the universality of the front; both are measured in the accompanying repository on six machines (the boundary columns $(01)^\infty$, $(0011)^\infty$, a coin and the constant 1, and the seed's column with two random initial states), on which the congruence holds in both regimes, twelve cases in all. Proposition 6.3 is a mechanism by which the agreement of all seeds up to the first genuine branching, recorded by Gravner and Griffeath [2, §8], can persist beyond it.

Computation 6.4 (EXACT). The edge machine of the accompanying repository evolves (1) on the rows $M_d - 64$ to $M_d + 66$ around the front, diagonal by diagonal, with the labels computed by Lemma 3.1 and one transient bit per diagonal, the cell $D_{d+1}(M_d - 64)$, entering 64 rows on the transient side of the front. The window determines the next diagonal on the same rows, and the new front is read where the new diagonal last differs from its label, so given the true bits the machine reproduces the seed's front exactly. A Python version run with the true bits does so with no deviation over 3,000 to 5,000 diagonals at three window depths, and the C version agrees with

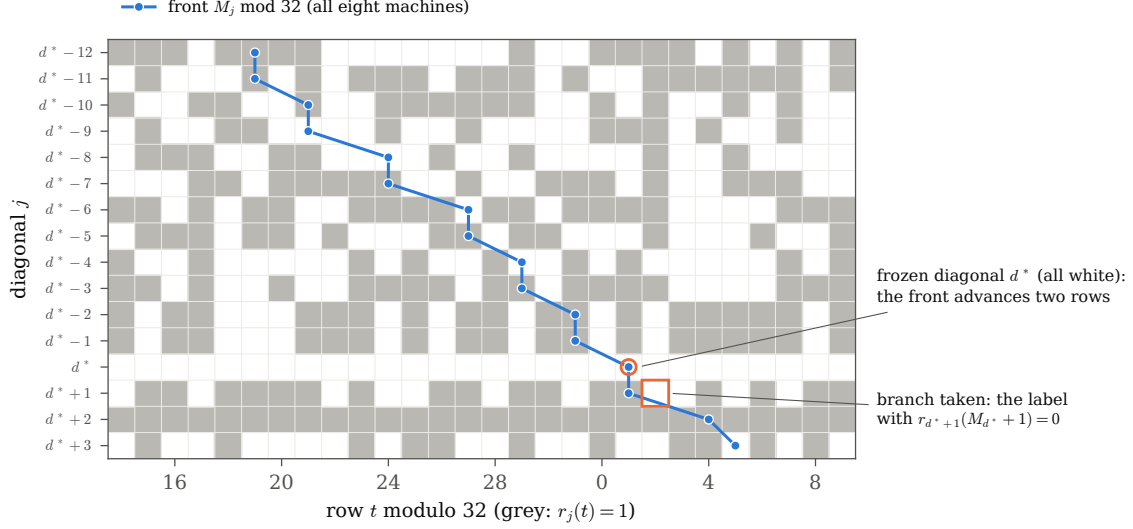


Figure 3: The front's path through the labels at the eighth frozen diagonal $d^* = 1,420,878,968$. Rows are diagonals j , columns are rows t modulo 32, grey cells are $r_j(t) = 1$. Each move from diagonal $j - 1$ to j lands on the first grey cell of row $j - 1$ after the previous front, as the landing rule requires. The frozen diagonal is all white; the front advances there from 31 to 1 (Theorem 6.2), and the branch taken is the candidate that is white in the cell after the new front (square).

it over 200,000 diagonals with constant input bits. We drove that bit with a fair coin. Twenty runs from a crystal start at $d = 90,000$ (the window filled with the labels, with no transient, and the front put at the seed's row $M_{90,000} = 120,085$) all reach d^* , all have $M_{d^*} \equiv 1$ modulo 32, and all take the branch **6cbda353**. Eight further runs, started $2 \cdot 10^6$ diagonals before d^* with start rows 7 apart, have one and the same front history modulo 32 over the sixteen diagonals $d^* - 12, \dots, d^* + 3$, although their absolute rows differ by up to 2,336 (Figure 3).

Theorem 6.5 (the seed's branch at d^* ; PROVED given the front's path). *Suppose that, as in each of the eight recorded machines of Computation 6.4, $M_{d^*-3} \equiv 29$ modulo 32, the front moves at $d^* - 2$, and $M_{d^*-1} = M_{d^*-2}$. Then the front advances at d^* , and the seed's branch is **6cbda353**, the one on which the period doubles at 2,107,985,255.*

Proof. Write $d = d^*$. The label cells used are $r_{d-3}(30) = 0$, $r_{d-3}(31) = 1$, $r_{d-2}(31) = 1$, $r_{d-1}(0) = 0$ and $r_{d-1}(1) = 1$ (rows modulo 32), read from $r_{d-3} = \mathbf{b792cb87}$ and $r_{d-2} = r_{d-1} = \mathbf{dae372fa}$. By the landing rule the move at $d - 2$ goes from M_{d-3} to the next black cell of r_{d-3} , which is $M_{d-3} + 2$. So $M_{d-2} = M_{d-1} = m$ with $m = M_{d-3} + 2 \equiv 31$.

1. $D_{d-2}(m) = \neg r_{d-2}(m) = 0$: the defect at the landing row of the move at $d - 2$ (Lemma 2.1(d)).
2. $D_{d-1}(m+1) = r_{d-1}(m+1) = r_{d-1}(0) = 0$, since row $m+1$ lies beyond the front $M_{d-1} = m$.
3. $D_{d-3}(m) = r_{d-3}(m) = 1$, since row m lies beyond the front $M_{d-3} = m - 2$.
4. By (1) on diagonal $d - 1$: $0 = D_{d-1}(m+1) = D_{d-3}(m) \oplus (D_{d-2}(m) \vee D_{d-1}(m)) = 1 \oplus D_{d-1}(m)$, so $D_{d-1}(m) = 1$.
5. By (1) on diagonal d : $D_d(m+1) = D_{d-2}(m) \oplus (D_{d-1}(m) \vee D_d(m)) = 0 \oplus 1 = 1$.

By Theorem 6.2 the front advances to the first black cell of r_{d-1} at or after row $m + 1 \equiv 0$, which is $m + 2 \equiv 1$, and the branch has $r_{d+1}(m + 3) = r_{d+1}(2) = 0$. Bit 2 of 6cbda353 is 0 and bit 2 of 93425cac is 1. $\square$

Remark 6.6 (the wall alone does not decide; EXACT). Take every front history of depth 8 before d^* that is consistent with the landing rule, from every starting phase. Propagating (1) as a constraint from the known cells (every label cell beyond its front, and the defect at each landing row), as in the proof of Theorem 6.5, forces an advance for 1,038 of the 4,096 histories, a stall for 1,791, and neither for 1,267. Both verdicts occur, at depth 5, at every frozen diagonal of the seed from 28 to 87,866. So at none of these frozen diagonals does the wall alone decide the branch: paths of the front that are all consistent with it force different branches. Nor does the path always decide it: at 28 and 58,286 the seed's own advance is not forced by this propagation at any depth up to 32, and there the transient decides. The hypothesis of Theorem 6.5 is a statement about the front's path, which is measured, not proved.

Remark 6.7 (the doubling itself; EXACT). Eight coin-driven machines, started $2 \cdot 10^6$ diagonals before 2,107,985,254, all stall there. Their branches split four to four, because the candidates b6e14797491eb868 and 491eb868b6e14797 (bit t of the 64-bit word is the cell on row t modulo 64) are translates by 32 rows and each machine picks by its front row modulo 64. Seven of the eight share one front history modulo 32, a move to row 8 modulo 32 five diagonals before the doubling and then stalls, a case Theorem 6.2 does not cover; the eighth is at row 26. Seen from the front, the seven take the same label, so which translate the seed takes is decided by its front row modulo 64, and no datum of the period-32 wall fixes that. So the absolute phase of the seed's period-64 wall is one bit the recursion cannot supply. Frozen diagonals, forks and everything measured relative to the front do not depend on it.

7 What the wall alone produces

The labels define a centre column of their own, $b_B(t) = r_t(t)$: the cell the stripes would put on the centre column if every diagonal were periodic from the start. Lemma 3.1 computes it without the rule, 10^9 bits in 35 seconds.

MEASURED: b_B passes every fair-coin test we applied to 10^9 bits but one. Its density is 0.500029, its run counts for lengths 1 to 12 are within 1.3 standard deviations of a coin's, and the χ^2 of its blocks of length 8, 12 and 16 lie 1.5, 0.3 and 0.2 standard deviations below a coin's mean. No context of up to 16 bits predicts it better than a coin does; that test was calibrated against a coin and against the real column at equal length. It has no period up to 10^7 on its last 10^8 bits. At the 26 lags we tested between 1 and 128 its autocorrelation is within 2.1 standard deviations of zero, except at lag 64, where it is 6.2 standard deviations below, a trace of the period-32 stripes; this is the one test it fails. On rows 90,001–1,130,000 it agrees with the real centre column c on a fraction 0.49973 ($z = -0.55$).

Proposition 7.1 (decomposition; PROVED, with a MEASURED reading). *$c(t) = b_B(t) \oplus \Delta_t(t)$ for every t . On rows 1,000–1,130,000 the transient bit $\Delta_t(t)$ behaves as a fair coin: density 0.50015, the 2×2 table of c against b_B flat to 10^{-3} , and every conditional frequency we measured within 2.1 standard deviations of $\frac{1}{2}$: given the wall's phase (t modulo 16, 32 or 64), the three wall cells around it, the distance to the next black cell of r_t , the period, whether the front moves at diagonal t , how far row t is from the front, and the mode of the defect there.*

The identity is immediate, since $D_t(t) = u_0^t = c(t)$. For every t from 20 to 800,000, $\tau_t > t$: row t lies in the transient of diagonal t , so $\Delta_t(t)$ is a live transient bit (the front row, $M_t \approx 1.32t$, is far

past it). That this persists for all t is, up to one row, the conjecture that $\tau_d \geq d$ for every $d \geq 20$ (Conjecture H of [1]; EXACT to $d = 4 \cdot 10^6$, OPEN beyond), which says that the stripes never cross the centre column. The centre column is therefore a sequence the wall computes, XOR a transient bit on which, as far as measured, nothing the stripes know has any influence.

8 The front: speed and stalls

MEASURED: in the coin-driven edge machine the front advances 1.321294 rows per diagonal on the seed's period-32 wall, the mean of twenty runs over $2.8 \cdot 10^{10}$ diagonals in all (spread between runs 0.000019, error of the mean 0.000004), and 1.321438 on the period-64 wall, four runs over $1.2 \cdot 10^{10}$ (spread 0.000007). On each block of 10^8 diagonals the twenty coins agree to within 0.0001 (standard deviation), while the twenty-run means of different blocks vary with standard deviation 0.00013, seven times what the coins alone would give, so the speed is a property of the wall. On period-16 walls it is about 1.333. The stall lengths have one geometric tail at both periods, with ratio 0.433 at period 32 and 0.434 at period 64. Leaving out the one stall at the start of each run, while the transient fills in from the crystal start, the longest stalls seen are 24 at period 32 and 26 at period 64, against $P - 1 = 31$ and 63. Extrapolating the tail, a stall of 32 is expected once per $2.7 \cdot 10^{12}$ diagonals and one of 64 once per 10^{24} . This is the margin of the empirical bound $k_i \leq P_i - 1$ on the length k_i of the i th run of consecutive stalls, P_i the period there, which holds on the seed exactly up to diagonal 800,000 [1]. In [1] we prove that $\tau_d \geq d$ for all $d \geq 20$ follows from the finite check $\tau_d \geq d$ for $20 \leq d \leq 36$ together with two inequalities on the front holding from diagonal 36 on: one on the stall runs, $k_i \leq \sigma_i - d_i - P_i + 1$ with σ_i the row and d_i the diagonal at which the run begins, which asks much less than $k_i \leq P_i - 1$ once the slack $\sigma_i - d_i$ is large (on the seed's data it grows like $0.32 d_i$, so a stall run would have to last about $0.32 d_i - P_i$ diagonals to violate it); and one on a second quantity, the cancellation depth of the defects, checked exactly to 800,000 there and not addressed here.

9 Open problems

Each of the following is OPEN.

1. *The front's path at d^* .* Prove that the seed's front has the path the coin-driven models show, $M_{d^*-3} \equiv 29$ modulo 32, a move, then a stall. This makes Theorem 6.5 unconditional. More generally, explain why fronts driven by independent coins keep one phase against the wall (Computation 6.4), although the wall alone does not fix the branch (Remark 6.6).
2. *The fork law.* Prove that the distance between consecutive labels is asymptotically binomial within each regime, or only that frozen diagonals occur at rate $2^{-P(1+o(1))}$.
3. *The transient bit.* Prove anything about $\Delta_t(t)$ jointly with $b_B(t)$. Since $c(t) = b_B(t) \oplus \Delta_t(t)$ and b_B is computable, every statistical question about the centre column is one about this pair.

Data and code

All code, data and logs are in the accompanying repository:

<https://github.com/nateberzai-svg/rule30-left-edge>

- The recursion: `scripts/wall_recursion.c` and `scripts/wall_recursion128.c`; logs, including the frontier run and the second machine's checkpoints, in `results/wall_recursion_logs`.

- The edge machine: `scripts/edge_machine.c`; runs in `results/edge_machine_runs`.
- The fork law: `scripts/label_distance.c`; logs in `results/fork_law`.
- Theorem 6.5 and Remarks 6.6–6.7: `scripts/stall_lookback_trace.py` and `scripts/stall_census.py`; labels and machine logs in `results/fork_mechanism`.
- Section 7: `scripts/frozen_column.py` and `scripts/frozen_column_autocorr.py`; logs in `results/frozen_column`.
- The figures: `paper/figures/make_figures.py`.
- `paper/verify_claims.py` recomputes every number stated here from the labels to 800,000 in `paper/data` and from the logs, and prints each claim with the value it finds.

References

- [1] N. Berzai, The centre column of Rule 30: excluded periods, decimation of the Duhamel class, and the front of the diagonal defects, manuscript, 2026; `paper/RULE30_PAPER.pdf` in the repository above.
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